Lecture 1

Local action, local equation

In *Theoretical Mechanics* a motion is a curve $q(t)$ that makes the action $\int L(q,\dot q)\,\mathrm{d}t$ stationary under variations with fixed endpoints. Now put an oscillator at every point x of space. Its displacement $\phi(t,x)$ depends on time and on the point. The coupling between neighbouring oscillators brings in the space derivatives of ϕ. Such a function of time and space is a field. Its action is an integral over a region of spacetime. This course asks what a classical field theory is. The first answer, developed in Lectures 1 to 3, is that a field theory is given by a local action.

Objectives. *We define fields, Lagrangian densities and actions on compact regions. We differentiate the action along a variation and keep the boundary term. We prove that stationary fields are exactly the solutions of the Euler–Lagrange equations.*

By the end of this lecture you will be able to:

- Compute the Euler–Lagrange expression of a first-order Lagrangian density.
- Derive the first-variation formula, with its boundary terms, on a region between two graphs.
- Prove that stationarity under compactly supported variations is equivalent to the Euler–Lagrange equations.
- Compute the boundary term for a variation that does not vanish on the boundary.

1.1 Fields and actions

Spacetime is $\mathbb{R}^{1,d}=\mathbb{R}\times\mathbb{R}^d$ with coordinates $x^0=t$ and $x=(x^1,\dots,x^d)$. Greek indices μ,ν run over $0,\dots,d$, spatial indices i,j over $1,\dots,d$, and component indices a,b over $1,\dots,N$. A repeated index is summed. The Minkowski metric is $\eta=\mathrm{diag}(1,-1,\dots,-1)$. Its inverse $\eta^{\mu\nu}$ has the same entries. We write $\partial^\mu=\eta^{\mu\nu}\partial_\nu$. The wave operator is

$$\Box=\partial_\mu\partial^\mu=\partial_t^2-\Delta,\qquad \Delta=\partial_1^2+\cdots+\partial_d^2.$$

A signal sent from the origin at the speed of light $c=1$ travels on the future light cone $|x|=t$.

We often replace space $\mathbb{R}^d$ by the torus $\mathbb{T}^d$, with $\mathbb{T}=\mathbb{R}/2\pi\mathbb{Z}$, so that space is compact. A function on $\mathbb{T}^d$ is a function on $\mathbb{R}^d$ that is 2π-periodic in every coordinate. Spacetime M is then $\mathbb{R}\times\mathbb{R}^d$ or $\mathbb{R}\times\mathbb{T}^d$. Both are oriented by the volume form

$$\mathrm{vol}=\mathrm{d}t\wedge\mathrm{d}x^1\wedge\cdots\wedge\mathrm{d}x^d,$$

and we write $\mathrm{d}^dx=\mathrm{d}x^1\cdots\mathrm{d}x^d$ for integrals over space.

Definition 1.1 (Field). A *field* with N components on M is a C^2 map

$$\phi\colon M\to\mathbb{R}^N,\qquad x\mapsto\big(\phi^1(x),\dots,\phi^N(x)\big).$$

A complex field is a field with two real components, its real and imaginary parts.

The Lagrangian of a field is a function of the point x, the value $\phi(x)$ and the first derivatives $\partial_\mu\phi(x)$. We write $u = (u^a)$ for a point of $\mathbb{R}^N$ and $v = (v^a_\mu)$ for a point of $\mathbb{R}^{N(d+1)}$, the place where the derivatives $\partial_\mu\phi^a$ will go.

Definition 1.2 (Lagrangian density)**.** A *Lagrangian density* is a C^2 function

$$L\colon M \times \mathbb{R}^N \times \mathbb{R}^{N(d+1)} \to \mathbb{R}, \qquad (x,u,v) \mapsto L(x,u,v).$$

For a field ϕ we write

$$L[\phi](x) = L\big(x, \phi(x), \partial\phi(x)\big),$$

where $\partial\phi(x)$ is the array of the derivatives $\partial_\mu\phi^a(x)$.

The density is *local*: its value at x depends only on ϕ and its first derivatives at x. The space of the arguments (x,u,v) is called the first jet space.

Definition 1.3 (Action on a region)**.** A *region* is a compact subset $D \subset M$ that is the closure of its interior. The *action* of a field ϕ on the region D is

$$S_D[\phi] = \int_D L[\phi] \operatorname{vol}.$$

The main example is the *slab*

$$D = [t_0, t_1] \times \mathbb{T}^d, \qquad t_0 < t_1,$$

which is compact because the torus is. An integral over $\mathbb{T}^d$ is taken over one period $[0, 2\pi]^d$. The boundary of the slab consists of the two *slices* $\{t_0\} \times \mathbb{T}^d$ and $\{t_1\} \times \mathbb{T}^d$.

For our running example take one component, $N = 1$, and the *Klein–Gordon density*

$$L = \frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{1}{2}m^2\phi^2, \tag{1.1}$$

with a constant $m \geq 0$. It is the oscillator Lagrangian $\frac{1}{2}\dot q^2 - \frac{1}{2}m^2q^2$ at each point, plus the coupling term $-\frac{1}{2}|\nabla\phi|^2$ between neighbouring points.

1.2 The first variation

As in mechanics, we compare a field with nearby fields and differentiate the action.

Definition 1.4 (Variation and first variation)**.** A *variation* is a C^1 map

$$h\colon M \to \mathbb{R}^N.$$

The *first variation* of the action of a field ϕ on a region D in the direction h is

$$\delta S_D[\phi]h = \frac{\mathrm{d}}{\mathrm{d}\varepsilon}\int_D L\big(x, \phi + \varepsilon h, \partial\phi + \varepsilon\partial h\big)\operatorname{vol}\Big|_{\varepsilon=0}.$$

The integrand needs only first derivatives, so the integral makes sense for the C^1 maps $\phi + \varepsilon h$. The first variation involves the partial derivatives of L in u and in v, which we name now.

Definition 1.5 (Momentum current and Euler–Lagrange expression)**.** For a field ϕ the *momentum current* has the components

$$p_a^\mu(x) = \frac{\partial L}{\partial v_\mu^a}\big(x, \phi(x), \partial\phi(x)\big),$$

and the *Euler–Lagrange expression* has the components

$$E_a[\phi](x) = \frac{\partial L}{\partial u^a}\big(x, \phi(x), \partial\phi(x)\big) - \partial_\mu p_a^\mu(x).$$

In the second term ∂_μ differentiates the composite function $x \mapsto p_a^\mu(x)$, so it also differentiates through $\phi(x)$ and $\partial\phi(x)$. Since L and ϕ are C^2, the current p_a^μ is C^1 and $E_a[\phi]$ is continuous. The time component $p_a^0 = \partial L/\partial(\partial_t \phi^a)$ is the field version of the canonical momentum $p = \partial L/\partial\dot{q}$.

Theoretical Mechanics wrote the Euler–Lagrange expressions as $\mathcal{E} = \dot{p} - \partial L/\partial q$. When there is no space ($d = 0$), this is the negative of the present expression. We use the sign of Olver, *Applications of Lie Groups to Differential Equations*, §4.1, so that the first variation below is $\int E_a h^a$ vol plus boundary terms, with no minus sign.

For the Klein–Gordon density (1.1) we get $p^\mu = \partial^\mu\phi$, so $p^0 = \phi_t$ and $p^i = -\partial_i\phi$. Hence

$$E[\phi] = -m^2\phi - \partial_\mu\partial^\mu\phi = -\big(\Box\phi + m^2\phi\big).$$

We import one result, differentiation under the integral sign. It is a special case of the theorem on parameter integrals in *Real and Complex Analysis*, Lecture 4: a continuous function on a compact set is bounded, so a constant is an integrable majorant.

Theorem 1.6 (Differentiation under the integral · *imported*)**.** *If $F(\varepsilon, x)$ and $\partial_\varepsilon F(\varepsilon, x)$ are continuous on $[-1, 1] \times D$ with D compact, then*

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\int_D F \,\mathrm{vol} = \int_D \partial_\varepsilon F \,\mathrm{vol}.$$

The boundary terms are written for a region between two graphs over the torus. For C^1 functions $\tau_0 < \tau_1$ on $\mathbb{T}^d$ we set

$$D = \{(t, x) \in \mathbb{R} \times \mathbb{T}^d : \ \tau_0(x) \le t \le \tau_1(x)\}. \tag{1.2}$$

This set is a region, and its boundary is the union of the two graphs. Constant $\tau_0 = t_0$ and $\tau_1 = t_1$ give the slab.

Proposition 1.7 (First variation)**.** *Let L be a Lagrangian density, ϕ a field and h a variation.*

(a) *For the region (1.2), with $V^\mu = p_a^\mu h^a$,*

$$\delta S_D[\phi]h = \int_D E_a[\phi]\, h^a \,\mathrm{vol} + \int_{\mathbb{T}^d} \big(V^0 - V^i\partial_i\tau_1\big)\big|_{t=\tau_1(x)}\,\mathrm{d}^d x \\ - \int_{\mathbb{T}^d} \big(V^0 - V^i\partial_i\tau_0\big)\big|_{t=\tau_0(x)}\,\mathrm{d}^d x. \tag{1.3}$$

(b) *For any region D, if the support of h lies in the interior of D, then*

$$\delta S_D[\phi]h = \int_D E_a[\phi]\, h^a \,\mathrm{vol}.$$

Proof. We differentiate under the integral, apply the product rule, and integrate one term coordinate by coordinate.

First, set $F(\varepsilon, x) = L(x, \phi + \varepsilon h, \partial\phi + \varepsilon\partial h)$. It is continuous, and by the chain rule

$$\partial_\varepsilon F(\varepsilon, x) = \frac{\partial L}{\partial u^a}(\dots)\, h^a + \frac{\partial L}{\partial v^a_\mu}(\dots)\, \partial_\mu h^a,$$

with the arguments of F in place of the dots. This is continuous, because L is C^2 and h is C^1. The imported theorem gives

$$\delta S_D[\phi]h = \int_D \partial_\varepsilon F(0, x)\,\mathrm{vol}, \qquad \partial_\varepsilon F(0, x) = \frac{\partial L}{\partial u^a}[\phi]\, h^a + p^\mu_a\, \partial_\mu h^a.$$

Next, the product rule gives $p^\mu_a \partial_\mu h^a = \partial_\mu V^\mu - (\partial_\mu p^\mu_a) h^a$, with $V^\mu = p^\mu_a h^a$. Hence

$$\delta S_D[\phi]h = \int_D E_a[\phi]\, h^a\,\mathrm{vol} + \int_D \partial_\mu V^\mu\,\mathrm{vol},$$

and the vector field V is C^1.

Finally, we integrate $\partial_\mu V^\mu$ over D by Fubini's theorem, first over t from $\tau_0(x)$ to $\tau_1(x)$. The fundamental theorem of calculus gives

$$\int_D \partial_t V^0\,\mathrm{vol} = \int_{\mathbb{T}^d} \big[V^0(\tau_1(x), x) - V^0(\tau_0(x), x)\big]\,\mathrm{d}^d x.$$

For a spatial index i, the function $G(\alpha, \beta, x) = \int_\alpha^\beta V^i(t, x)\,\mathrm{d}t$ has the partial derivatives $V^i(\beta, x)$, $-V^i(\alpha, x)$ and $\int_\alpha^\beta \partial_i V^i\,\mathrm{d}t$. The chain rule applied to $x \mapsto G(\tau_0(x), \tau_1(x), x)$ gives

$$\int_{\tau_0}^{\tau_1} \partial_i V^i\,\mathrm{d}t = \partial_i \int_{\tau_0}^{\tau_1} V^i\,\mathrm{d}t - V^i(\tau_1, x)\,\partial_i\tau_1 + V^i(\tau_0, x)\,\partial_i\tau_0.$$

The first term on the right is the derivative of a 2π-periodic function, so its integral over a period of x^i vanishes. Adding the time and spatial terms gives the boundary terms of (1.3), which proves part (a).

In part (b), the support of h is a closed subset of the compact set D, so it is compact. The field V vanishes outside it, so $\int_D \partial_\mu V^\mu\,\mathrm{vol} = \int_M \partial_\mu V^\mu\,\mathrm{vol}$. Along every coordinate line, V either vanishes outside a bounded interval or is periodic. Hence each term $\int_M \partial_\mu V^\mu\,\mathrm{vol}$ is zero by the fundamental theorem of calculus, which proves part (b). ∎

On the slab $D = [t_0, t_1] \times \mathbb{T}^d$ the boundary terms become

$$\int_{\mathbb{T}^d} p^0_a h^a\,\mathrm{d}^d x\Big|_{t=t_1} - \int_{\mathbb{T}^d} p^0_a h^a\,\mathrm{d}^d x\Big|_{t=t_0}. \tag{1.4}$$

They are the field version of the boundary term $\big[p\,\delta q\big]_{t_0}^{t_1}$ of mechanics. In Figure 1.1 the vertical edges $x = 0$ and $x = 2\pi$ are identified, so the slab over the circle has no spatial boundary. The only boundary is the pair of slices, with the outward normal $+\partial_t$ on the top slice and $-\partial_t$ on the bottom one.

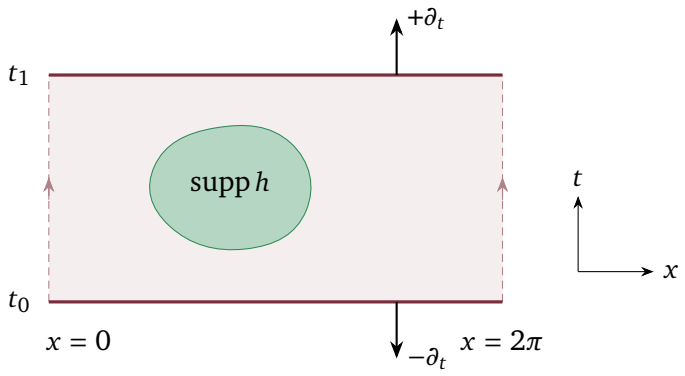


Figure 1.1: The slab $[t_0, t_1] \times \mathbb{T}$ with its two boundary slices (thick) and their outward normals. The shaded set is the support of a variation h that vanishes near the slices.

Remark 1.8 (The covariant boundary term)**.** Let $\mathrm{d}\Sigma_\mu = \iota_{\partial_\mu}\mathrm{vol}$, where ι inserts a vector into the first slot of a form. Thus $\mathrm{d}\Sigma_0 = \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$, and each $\mathrm{d}\Sigma_i$ contains the factor $\mathrm{d}t$. Along a graph $t = \tau(x)$ we have $\mathrm{d}t = \partial_j\tau\,\mathrm{d}x^j$, so $\mathrm{d}\Sigma_i$ pulls back to $-\partial_i\tau\,\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$. Hence $V^\mu \mathrm{d}\Sigma_\mu$ pulls back to $(V^0 - V^i\partial_i\tau)\,\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$ on either graph. Orient ∂D with the outward normal first, as in *Topology and Geometry*, Lecture 12. This orientation agrees with $\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^d$ on the top graph and is opposite on the bottom graph. So the boundary terms of (1.3) are $\int_{\partial D} V^\mu \mathrm{d}\Sigma_\mu$. Lecture 8 proves the analogous formula on $\mathbb{R}^d$ for slices that are not flat.

1.3 Stationary fields and the Euler–Lagrange equations

The slice terms in (1.4) are the reason to restrict the variations, exactly as Hamilton's principle in mechanics fixes the endpoints.

Definition 1.9 (Admissible variation, stationary field)**.** A variation h is *admissible on a region* D if its support lies in the interior of D. A field ϕ is *stationary* if $\delta S_D[\phi]h = 0$ for every region $D \subset M$ and every variation h admissible on D.

For an admissible variation, part (b) of Proposition 1.7 leaves only the integral of $E_a[\phi]h^a$. The following lemma shows that the vanishing of this integral for every admissible h forces $E[\phi] = 0$.

Lemma 1.10 (Fundamental lemma)**.** *Let $U \subset M$ be open and $g : U \to \mathbb{R}$ continuous. Suppose that*

$$\int_U g\,h\,\mathrm{vol} = 0 \quad \text{for every } C^2 \text{ function } h \text{ with compact support in } U.$$

Then $g = 0$ on U.

Proof. We argue by contradiction. Suppose $g(z) > 0$ at some point $z \in U$. We use coordinates $y = (y^0, \dots, y^d)$ centred at z, with the spatial coordinates in $(-\pi, \pi)^d$ when space is the torus, and write $|y|$ for the Euclidean length of y. By continuity there is $r \in (0, \pi)$ such that the closed ball $\bar{B}$, with $B = \{|y| < r\}$, lies in U and $g > \tfrac{1}{2}g(z)$ on $\bar{B}$. Take the bump

$$h(y) = \begin{cases} \big(r^2 - |y|^2\big)^3, & |y| < r, \\ 0, & \text{otherwise.} \end{cases}$$

It is C^2, because the cube vanishes to order three on the sphere $|y| = r$. It is positive on B. Its support is $\bar{B} \subset U$. Then

$$\int_U g\, h \,\mathrm{vol} \geq \tfrac{1}{2} g(z) \int_B h \,\mathrm{vol} > 0,$$

a contradiction. If $g(z) < 0$, apply the same argument to $-g$. ∎

Theorem 1.11 (Stationary fields). *A field ϕ is stationary if and only if it solves the* Euler–Lagrange equations

$$E_a[\phi] = \frac{\partial L}{\partial u^a}[\phi] - \partial_\mu p_a^\mu = 0 \qquad \text{on } M, \quad a = 1, \dots, N.$$

Proof. First suppose $E[\phi] = 0$. For a region D and a variation h admissible on D, part (b) of Proposition 1.7 gives $\delta S_D[\phi]h = \int_D E_a[\phi] h^a \,\mathrm{vol} = 0$.

Conversely, suppose ϕ is stationary, and fix a component a and a point $z \in M$. Let D be a closed coordinate ball about z of radius less than π, as in the proof of Lemma 1.10, and let U be its interior. For every C^2 function g with compact support in U, the variation h with $h^a = g$ and the other components zero is admissible on D. By part (b) of Proposition 1.7, $0 = \delta S_D[\phi]h = \int_U E_a[\phi]\, g \,\mathrm{vol}$. Lemma 1.10 gives $E_a[\phi] = 0$ on U, in particular at z. ∎

The theorem does not give existence. Lecture 4 constructs solutions of the Klein–Gordon equation on $\mathbb{R}^d$ from finite-energy initial data.

1.4 The Klein–Gordon field on a circle

We find the field equation of the density (1.1) and one of its solutions, and compute its energy. The data are

$$M = \mathbb{R} \times \mathbb{T}, \qquad L = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}m^2\phi^2, \qquad m = 3.$$

The momentum current is $p^0 = \phi_t$ and $p^1 = -\phi_x$. The remaining derivative is $\partial L/\partial u = -m^2\phi$. Hence

$$E[\phi] = -m^2\phi - \partial_t\phi_t - \partial_x(-\phi_x) = -\big(\phi_{tt} - \phi_{xx} + m^2\phi\big).$$

By Theorem 1.11 the stationary fields are the solutions of the *Klein–Gordon equation*

$$\phi_{tt} - \phi_{xx} + m^2\phi = 0.$$

We look for a solution of the form

$$\phi(t, x) = a(t) \cos 2x.$$

Its zeros in x stay fixed while its amplitude $a(t)$ oscillates. Such a solution is called a standing wave. The function $\cos 2x$ is 2π-periodic. Since $\phi_{xx} = -4a\cos 2x$, the equation becomes

$$\big(\ddot{a} + 4a + 9a\big)\cos 2x = 0.$$

Since $\cos 2x$ is not identically zero, this holds exactly when $\ddot{a} + 13a = 0$. This is an oscillator with frequency $\nu = \sqrt{13}$. With $a(0) = 1$ and $\dot{a}(0) = 0$ we get

$$\phi(t, x) = \cos\big(\sqrt{13}\, t\big)\cos 2x,$$

with period $2\pi/\sqrt{13} \approx 1.743$. The mode with k waves around the circle oscillates with $\nu^2 = k^2 + m^2$. The coupling term contributes k^2. The mass term contributes m^2.

The energy of the field adds the kinetic term, the coupling term and the mass term of (1.1), all with the plus sign:

$$E_{\text{en}} = \int_{\mathbb{T}} \Big(\frac{1}{2}\phi_t^2 + \frac{1}{2}\phi_x^2 + \frac{1}{2}m^2\phi^2 \Big) \mathrm{d}x.$$

For the standing wave we write $s = \sin(\sqrt{13}\,t)$ and $c = \cos(\sqrt{13}\,t)$, and use $\int_{\mathbb{T}} \cos^2 2x \,\mathrm{d}x = \int_{\mathbb{T}} \sin^2 2x \,\mathrm{d}x = \pi$. The three terms give

$$E_{\text{en}} = \frac{13\pi}{2}s^2 + 2\pi c^2 + \frac{9\pi}{2}c^2 = \frac{13\pi}{2}$$

at every time. Lecture 3 derives this energy from time translations and proves that it is conserved for every solution.

1.5 A boundary term in the first variation

Theorem 1.11 uses only variations that vanish near the boundary of a region. The restriction is necessary. A solution of the Euler–Lagrange equations can have a nonzero first variation in a direction that moves its boundary values.

Take the massless field, $m = 0$, on the slab $D = [0, T] \times \mathbb{T}$ with $T > 0$, and the field $\phi(t, x) = t$. It solves $\phi_{tt} - \phi_{xx} = 0$, so $E[\phi] = 0$. Take the variation $h(t, x) = t/T$, with $h = 0$ at $t = 0$ and $h = 1$ at $t = T$. It is not admissible. By (1.3) and (1.4), with $p^0 = \phi_t = 1$,

$$\delta S_D[\phi]h = 0 + \int_{\mathbb{T}} 1 \cdot 1 \,\mathrm{d}x - \int_{\mathbb{T}} 1 \cdot 0 \,\mathrm{d}x = 2\pi.$$

A direct computation agrees:

$$S_D[\phi + \varepsilon h] = \int_0^T \int_{\mathbb{T}} \frac{1}{2}\Big(1 + \frac{\varepsilon}{T}\Big)^2 \mathrm{d}x \,\mathrm{d}t = \pi T \Big(1 + \frac{\varepsilon}{T}\Big)^2,$$

whose derivative at $\varepsilon = 0$ is 2π.

So the first variation of a solution need not vanish in a direction that moves the boundary values. Lecture 8 builds the symplectic form on the space of solutions out of this boundary term. Lecture 9 must control the same term when the field is the electromagnetic potential.

Summary. The action of a field on a compact region is the integral of a local density $L(x, \phi, \partial\phi)$. On a region between two graphs, its first variation is $\int_D E_a[\phi]h^a$ vol plus boundary terms built from the momentum current. A field is stationary exactly when $E[\phi] = 0$. For the Klein–Gordon density this is $\Box\phi + m^2\phi = 0$. Lecture 2 turns symmetries into conservation laws with the same formula.

Self-check. Find the Euler–Lagrange equation of the density

$$L = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi)$$

for a C^2 function $V\colon \mathbb{R} \to \mathbb{R}$.

Answer. Here $p^\mu = \partial^\mu\phi$ and $\partial L/\partial u = -V'(\phi)$, so $E[\phi] = -V'(\phi) - \Box\phi$. The equation is $\Box\phi + V'(\phi) = 0$.

Self-check. In the example of Section 1.5, replace h by $h(t,x) = \sin(\pi t/T)$. Is h admissible on $D = [0,T] \times \mathbb{T}$, and what is $\delta S_D[\phi]h$?

Answer. It is not admissible: its support is all of M, which does not lie in the interior of D. Still, h vanishes on both slices, so the slice terms (1.4) are zero. Since $E[\phi] = 0$, we get $\delta S_D[\phi]h = 0$.

Self-check. In the same example take the constant variation $h = 1$. Compute the slice terms (1.4), and check the result against $S_D[\phi + \varepsilon h]$.

Answer. The slice terms are $\int_{\mathbb{T}} 1\,\mathrm{d}x - \int_{\mathbb{T}} 1\,\mathrm{d}x = 2\pi - 2\pi = 0$. Directly, $\phi + \varepsilon$ has the same derivatives as ϕ, so $S_D[\phi + \varepsilon h]$ does not depend on ε, and its derivative is 0.